

SARAH MACGILLIVRAY

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Professional Summary

Dynamic Embedded Systems Engineer with a proven track record at Gluxkind, Inc., specializing in C++ firmware development and multithreaded programming. Achieved a 75% reduction in debugging cycles through innovative diagnostic solutions. Adept at collaborating in teams and leveraging machine learning techniques to enhance product reliability and efficiency.

Skills

- C/C++ and Java
- Embedded systems development
- Python and Swift
- ARM assembly language
- Visual Studio Code and IntelliJ IDEA
- MATLAB and Arduino IDE
- GitHub collaboration
- I2C/SPI interfacing
- Client-server architecture
- Test generation and debugging
- Multithreaded programming
- SystemVerilog
- FPGA configuration
- Linux Kernel Development
- Firmware development for robotics
- Machine learning techniques and evaluation
- Data analysis methodologies
- FreeRTOS systems
- Team collaboration
- Problem solving
- ModelSim and Quartus
- Circuit analysis and PCB design
- Altium design
- Waveform generation and oscilloscope usage
- TraxMaker for hardware testing

Experience

- Embedded Systems Engineer Co-op**
Gluxkind, Inc.

05/2025 to 09/2025
Vancouver, Canada

 - Architected high-performance, multi-threaded audio streaming engine in C++ on FreeRTOS, achieving sub-10ms audio latency for embedded therapeutic audio product.
 - Developed MP3 decoder integration with precise VBR frame analysis and ID3v2.4 metadata extraction, ensuring seamless playback across formats.
 - Built real-time diagnostic platform with optimized logging macros and web dashboard, reducing internal debugging cycles by 75%.
 - Engineered fault-tolerant inter-process communication system using FreeRTOS message queues, ensuring reliable audio delivery under all conditions.
 - Created full-stack testing framework including custom Bluetooth player website, enabling comprehensive hardware-software integration validation.
 - Led debugging initiatives for real-time firmware issues, systematically diagnosing user-reported bugs to enhance product reliability.
- Electrical Engineering Co-op**
DarkVision

05/2024 to 12/2024
Vancouver, Canada

 - Programmed firmware for ultrasound hardware using C++ and Python, collaborating with software teams through GitHub PR review processes to deliver functional and efficient production code.
 - Implemented SPI communication protocols to interface with underground ultrasound technology, enabling reliable data transmission and hardware control for downhole applications.
 - Designed and produced PCBs using Altium Designer schematic and layout tools, meeting engineering requirements for space-efficient and reliable electronics in harsh industrial environments.

- Led R&D automation project creating HTML interface for PO and quoting processes, achieving 90% reduction in quoting time and significantly improving operational efficiency.

Image Processing Technician **TechInsights**

05/2023 to 09/2023
Ottawa, Canada

- Analyzed semiconductor images at 4nm resolution to create virtualized circuit representations, working with cutting-edge chips including Apple A16 Bionic, Samsung, and SK Hynix memory devices.
- Trained machine learning algorithms for automated image correction and transistor extraction, reducing product delivery timelines by multiple days through improved analysis efficiency.
- Performed signal tracing on multilayer circuits to identify and correct breaks and shorts, ensuring accurate reverse engineering results.
- Generated composite GDS-II files by stitching semiconductor layer images (vias, metals, polysilicon) for comprehensive circuit documentation.

Education

Bachelor of Applied Science: Computer Engineering
University of British Columbia
GPA: 85%

Expected in 05/2027

Technical Projects

Client-Server IoT System

- Architected multithreaded server infrastructure connecting multiple clients to sensors and actuators using concurrent programming and real-time event handling.
- Implemented robust request-response protocols ensuring reliable communication and comprehensive data logging.
- Achieved 95%+ test coverage through thorough unit testing and edge case analysis.
- Led collaborative development on GitHub, organizing team meetings and defining development milestones.

Autonomous Competition Robots

- Designed and built competition-winning robots: firefighter robot (first to complete maze and extinguish flame) and sumo robot (competition winner).
- Created circuit schematics and space-optimized PCB layouts, showcasing hardware design and manufacturing proficiency.
- Programmed PIC40 microcontrollers in Basic for autonomous navigation, sensor processing, and actuator control.
- Performed systematic troubleshooting with multimeter analysis to identify and resolve hardware issues.

Activities and Honors

- UBC Varsity Cross Country and Track Athlete and Team Captain 2022-Present

Awards

- Dean's Honors List (2022-present)
- Academic All Canadian (2022-present)
- Academic All American (2022-present)
- Women in Engineering Award (2023)